

Modeling Discrete Linear Change and Arithmetic Sums

Answer Key

Algebra 1

Quick Answers

- | | |
|--|---|
| 1. height of a stack of 6 = 127 cm | 8. cost of a 7-hour call-out = 455 |
| 2. total paid after 8 months = 206 | 9. total bricks in 12 rows = 378 |
| 3. total minutes in 8 weeks = 208 min | 10. the month the allowance reaches 3000 = 11 |
| 4. the day the tank is empty = 16 | 11. (a) the year the two offers match = 3, (b) |
| 5. seats in row 15 = 62 | six-year total from the first offer = 36900 |
| 6. seats in rows 1 through 15 = 615 | 12. (a) seats in the last row = 64, (b) total seats |
| 7. the week the balance reaches 261 = 13 | in the hall = 984, — |
-

What is verified

11 of 12 problems fully machine-checked against SymPy.
 — marks an answer only you can judge — problem 12. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1
 HSF-LE.A.2 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 15 tagged checks.

Common wrong answers

1. *If they got 136: added the extra height six times instead of five*
 5. *If they got 65: multiplied the step by the row number instead of one less*
 9. *If they got 346.5: averaged the first and last rows but multiplied by the number of gaps instead of the number of rows*
-

1. Stack of chairs: first chair 82 cm, each extra chair adds 9 cm. Height of a stack of 6?
Solution: Let n count chairs, so $a_n = 82 + 9(n - 1)$. A stack of six has five *added* chairs:
 $82 + 5(9) = 82 + 45$. 127 cm
-

2. Gym: \$30 to join plus \$22 a month. Total paid after 8 months?

Solution: Here n counts months and the joining fee is paid once, so the total is $30 + 22n$.
At $n = 8$: $30 + 176$. 206

3. Running: 12 minutes in week 1, +4 minutes each week. Total minutes over 8 weeks?
Solution: Weekly minutes form the sequence $a_n = 12 + 4(n - 1)$, so week 8 is $12 + 7(4) = 40$. Pair first with last: $S_8 = \frac{8}{2}(12 + 40) = 4(52)$. 208 min

4. Tank: 240 litres, losing 15 litres a day. On which day is it empty?
Solution: Empty means the amount left is zero: $240 - 15n = 0$, so $15n = 240$. Check: $240 - 15(16) = 0$ ✓ n = 16

5. Theatre: 20 seats in row 1, +3 per row. Seats in row 15?
Solution: Row 15 is fourteen rows past row 1, not fifteen: $20 + 14(3) = 20 + 42$. 62

6. Same theatre. Total seats in rows 1 through 15?
Solution: The first and last rows hold 20 and 62 seats, and there are 15 rows, so $S_{15} = \frac{15}{2}(20 + 62) = \frac{15}{2}(82) = 15(41)$. 615

7. Savings: \$45 after week 1, +\$18 a week. Which week reaches \$261?
Solution: Solve $45 + 18(n - 1) = 261$. Subtract 45: $18(n - 1) = 216$, so $n - 1 = 12$. Check: $45 + 18(12) = 261$ ✓ n = 13

8. Plumber: \$95 for the first hour, \$60 for each additional hour. Cost of a 7-hour call-out?
Solution: With n counting hours on site, $a_n = 95 + 60(n - 1)$; six of the seven hours are charged at the lower rate: $95 + 6(60) = 95 + 360$. 455

9. Brick pile: 48 in the bottom row, 3 fewer each row up, 12 rows. Total bricks?
Solution: With n counting rows from the bottom, $a_n = 48 - 3(n - 1)$, so the top row holds $48 - 11(3) = 15$. Twelve rows, so $S_{12} = \frac{12}{2}(48 + 15) = 6(63)$. 378

10. Data plan: 500 MB in month 1, +250 MB a month. Which month first reaches 3000 MB?
Solution: Solve $500 + 250(n - 1) = 3000$. Subtract 500: $250(n - 1) = 2500$, so $n - 1 = 10$. Check: $500 + 250(10) = 3000$ ✓ n = 11

11. Offer A: \$2400 in year 1, +\$1500 a year. Offer B: \$3600 in year 1, +\$900 a year. (a) Equal year. (b) Six-year total for A.

Solution (a): Set the two year- n salaries equal: $2400 + 1500(n - 1) = 3600 + 900(n - 1)$. Collect: $600(n - 1) = 1200$, so $n - 1 = 2$. Check: A pays $2400 + 3000 = 5400$ and B pays $3600 + 1800 = 5400$ ✓ $n = 3$

Solution (b): Offer A in year 6 pays $2400 + 5(1500) = 9900$, so the six-year total is $S_6 = \frac{6}{2}(2400 + 9900) = 3(12300)$. 36900

12. Hall of 24 rows: 18 seats in row 1, +2 per row. (a) Last row. (b) Total. (c) Why doubling the rows more than doubles capacity.

Solution (a): Row 24 is twenty-three rows past row 1: $18 + 23(2) = 18 + 46$. 64

Solution (b): $S_{24} = \frac{24}{2}(18 + 64) = 12(82)$. 984

(c) — instructor-judged. A correct answer says the rows added on top are all larger than the rows already there, because each row holds two more seats than the one below, so the extra twenty-four rows contribute more than the original twenty-four did and the total more than doubles. (For reference, $S_{48} = \frac{48}{2}(18 + 112) = 3120$, well past 2×984 .) Full credit for tying the argument to the rows growing; partial credit for computing the doubled total with no reason.